THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $\triangle ABC$ right-angled at B, $AB = 5$, $BC = 12$. If $BD \perp AC$, then AD (in cm) is: 1

(a) $\frac{144}{13}$ **(b)** $\frac{5}{2}$ **(c)** $\frac{25}{13}$ **(d)** $\frac{60}{13}$

2. A square has diagonal $7\sqrt{2}$ cm. Its area (cm²) is: 1

(a) 98 **(b)** 196 **(c)** 49 **(d)** 14

3. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 3 : 4$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 9: 1

(a) 40 **(b)** 16 **(c)** 58 **(d)** 49

4. In a right triangle with hypotenuse $c = 17$ and one leg $a = 8$, the area (sq. units) is: 1

(a) 120 **(b)** 60 **(c)** $\frac{255}{2}$ **(d)** 68

5. The altitude on the hypotenuse of the right triangle with legs 8 and 15 is of length (cm): 1

(a) $\frac{17}{2}$ **(b)** $\frac{120}{17}$ **(c)** 60 **(d)** $\frac{23}{2}$

6. If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 20$ and $ar(PQR) = 605$ and $AB = 4$, then $PQ =$ 1

(a) 24 **(b)** 4 **(c)** 22 **(d)** 242

7. In $\triangle ABC$ right-angled at B, $AB = 8$, $BC = 15$. If $BD \perp AC$, then AD (in cm) is: 1

(a) $\frac{64}{17}$ **(b)** 4 **(c)** $\frac{120}{17}$ **(d)** $\frac{225}{17}$

8. A square has diagonal $9\sqrt{2}$ cm. Its area (cm²) is: 1

(a) 18 **(b)** 324 **(c)** 81 **(d)** 162

9. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 1 : 4$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 1: 1

(a) 26 **(b)** 24 **(c)** 25 **(d)** 16

10. In a right triangle with hypotenuse $c = 25$ and one leg $a = 7$, the area (sq. units) is: 1

(a) 168 **(b)** $\frac{175}{2}$ **(c)** 84 **(d)** 300

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The areas of two similar triangles are 32 cm² and 162 cm². If a side of the first is 4 cm, find the corresponding side of the second. 2

12. Find the altitude on the hypotenuse of the right triangle with legs 5 cm and 12 cm. 2

13. In $\triangle ABC$, $DE \parallel BC$, $AD = 4$ cm, $AB = 9$ cm and $BC = 18$ cm. Find DE. 2

14. The areas of two similar triangles are 20 cm² and 245 cm². If a side of the first is 2 cm, find the corresponding side of the second. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 5$ cm and $DC = 20$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

16. Find the altitude of an equilateral triangle of side 10 cm, in terms of $\sqrt{3}$. 3

17. $\triangle ABC$ is right-angled at B with $AB = 8$ cm and $BC = 15$ cm; $BD \perp AC$. Find AD, DC and BD. 3

OR

In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 6$ cm and $DC = 21$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

18. Find the altitude of an equilateral triangle of side 12 cm, in terms of $\sqrt{3}$. 3

OR

$\triangle ABC$ is right-angled at B with $AB = 7$ cm and $BC = 24$ cm; $BD \perp AC$. Find AD, DC and BD. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two ships leave a port at the same time; one sails 48 km due north and the other 90 km due west. Find the distance between them and the ratio of this distance to the one if both had sailed only half as far. 5

OR

A girl 2 m tall walks away from a lamp post 6 m high at 1 m/s. Find the length of her shadow after 8 seconds. 5

20. Two ships leave a port at the same time; one sails 90 km due north and the other 120 km due west. Find the distance between them and the ratio of this distance to the one if both had sailed only half as far. 5

OR

A girl 3 m tall walks away from a lamp post 9 m high at 2 m/s. Find the length of her shadow after 6 seconds. 5

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND